

MICROWAVE TEST BENCH (KLYSTRON BASED)



SCOPE

The MI2000 series of Microwave Test Benches are precision made microwave systems, which use standard type rectangular wave-guide components to illustrate the essential elements of this field for study. These Training Benches are completely self-contained & provided with the means to allow students to carry out practical work at extremely low cost. A comprehensive manual containing extensive microwave theory and a progressive series of assignments is supplied with the Trainer.

Features:

- A selection of wave-guide components
- The power supply for microwave source
- A meter, which monitors the detector output
- Wall Chart Experiment

Experiments

- Study the characteristics of klystron Tube and its tuning range.
- To determine the frequency & wavelength in a rectangular waveguide working TE₁₀ mode
- To study unknown impedance with smith chart
- To study the square law behavior of a microwave crystal detector
- To study the variable attenuator
- To study the square law behavior of a microwave crystal detector
- To study the E plane TEE, H Plane TEE
- To study MHD Coupler

VSWR Meter	: 1
Klystron Power Supply	: 1
Diode Detector with Precision Mount	: 1
Variable Attenuator 20dB	: 1
Frequency Meter, DR	: 1
Reflex Klystron Oscillator with Precision mount	: 1
Isolator	: 1
Matched Termination	: 1
Cooling Fan	: 1
ACCESSORIES	:
Wave Guide section 15 cm	2
Wave Guide Stands	4
BNC to BNC Cable	2
N to BNC Cable	1
Operating manual	1
Mains Cord	2
Additional Accessories (As per Requirement)	:
Pyramidal Horn Antenna	: 2
Standard Gain Horn Antenna	2
Parabolic Dish Antenna	: 2
E Plane Sectoral Horn	2
H Plane Sectoral Horn	2
Dielectric Antenna	2
Radiation Pattern Turn Table Gear Box type	1
90 degree bend	1
Variable short	1
Matched Load	1
S S Tuner	1
Fixed Short	1
Cooling Fan	1



Note : Specifications can be altered without notice in our constant efforts for improvement.

MICROLINE INDIA

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www.microlineindia.in

MI- KPS151 KLYSTRON POWER SUPPLY



Klystron Power is a state of art solid state regulated Power supply for operating low power Klystron. It incorporates number of proprietary features.

Features:

- Regulated Beam supply and Repeller supply voltages
- LED Digital meter for Beam voltage, current and Repeller voltage
- Modular construction for easy maintenance
- In addition to AM and FM modulation of beam current, provisions for external modulation have been provided

Beam Supply :

- Voltage 200-450 VDC, Variable
- Current 50mA
- Regulation 0.25% to 10% I/P Variation Ripple : < 5m Vrms (On Load)
- Repeller Supply : -10V to -270V DC, Regulation .25% to 10% I/P variation
- Filament supply : 6.3VDC (adjustable at rear panel)
- Overload Trip current : 50mA

Display : Digital display for

1. Beam Voltage
2. Beam current
3. Repeller voltage

External Modulation Selector : CW / AM / FM

Frequency Range: AM-500Hz - 2500Hz, FM-50-300Hz

Amplitude: 0-110 Vpp

Digital Display : Beam Voltage (V), Current (I), (Repeller)

Stand by Mode: Only heater supply ON

Connectors : 8-pin octal socket BNC F for external modulation

: 2-pin socket for cooling fan

: BNC F for saw tooth monitoring

Power Supply :220V $\pm$ 10%,50Hz

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Display : ANALOG

Sensitivity: 0.1 mV for 200 ohm input impedance

Noise Level : < 0.02V

Attenuation Range : 0 - 60dB in 10dB steps

Input impedance selector: low crystal based 200ohm, high crystal based
200Kohm, Bolo 200, 4.5mA bias; Bolo 200, 8.7mA bias

Display: VSWR Scale, dB Scale

Meter Scales: SWR 1-4, SWR 3-10 (Expanded), SWR 1-1.3, dB 0-10,
expanded 0-2 dB

Amplifier type: High gain tuned at 1000Hz

Modulation Frequency

Modes: Normal / Expanded

Gain Control: Course adjustment 10dB minimum, Fine adjustment 2dB minimum

Recorder output level : 1V for full scale deflection of meter

Input Connector: BNC (F)

Input Frequency : 1 KHz $\pm$ 10%

Power: 230 Volts AC

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Isolator

- VSWR: ≤ 1.15
- Insertion Loss: ≤ 0.46 dB
- Isolation: ≥ 20 dB
- Frequency Meter
- Accuracy: $\pm 2\%$
- Calibration Increment: 5 MHz

Slotted Section

- Residual VSWR: 1.01
- Slope: ± 0.2 dB
- Variable Attenuator
- Range: 20 dB

Matched Termination

- VSWR: 1.03
- Return Loss: -33 dB

Detector Mount

- Detector Type: IN21 or equivalent
- Output: BNC Connector

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